25901177

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) REGULAR EXAMINATION March 2026

Fourth SEMESTER

Discipline Specific Core Course (DSC) - MG4DSCCMA200 - DATA ANALYTICS USING R

(2024 ADMISSION ONWARDS)

Duration: 1 hour 30 minutes **Maximum Marks: 50**

SCHEME FOR VALUATION

PART A

Very Short Answer Questions Answer all questions

Each question carries 2 marks

1. **Define the repeat loop in R.** **[U] / [CO1]**
   Infinite loop. Repeats code until break statement
2. **Explain the purpose of the dir() function in R.** **[K] / [CO2]**
   Lists files. Shows directory contents
3. **Write any two advantages of a 3D pie chart over the bar chart.** **[U] / [CO3]**
   Shows proportions clearly , More visual and attractive
4. **Recall the file extension of an Excel file?** **[K] / [CO2]**
   .xlsx
5. **Write the output of the code: data <- c(10, 20, 30, 40, 50) mean(data).** **[A] / [CO3]**
   Output: 30

[2x5 = 10]

Part B

Short Answer Questions Answer any 5 questions

Each question carries 5 marks

6. **Compare the if–else if ladder and switch statement (any five points).[U] / [CO1]**
   (***3 Marks – Syntax of if–else if ladder and switch statement , 2 Marks – Any five comparison points)***
   if–else – Checks conditions one by one, Used for ranges/conditions, More flexible

   switch – Matches value with cases, Used for fixed options, more efficient
7. **Explain the concept of melting a data frame. [K] / [CO2]**
   (3 Marks – Definition,2 Marks – Example)

Converts wide format to long format, combines multiple columns, creates variable and value columns, used for data reshaping, Helps in analysis

8. **Compare a scatter plot, a whisker plot, and a box plot.** **[U] / [CO3]**
   (3 Marks – Definition and comparison of Scatter Plot, Whisker Plot, and Box Plot 2 Marks – Example or diagram of the plots)

   Scatter plot – Relationship between variables, Two numerical variables

   Box plot – Shows distribution

   Whisker plot – Shows spread of data

   Box/Whisker – Summary statistics

9. **Explain vectors in R with suitable examples.** **[U] / [CO1]**

   (3 Marks for vector and its definition ,2 marks for example)

   Collection of elements

   Same data type

   Created using c() function

   Example: x <- c(1, 2, 3)

   Used to store data

10. **Create a data frame product with ProductID, ProductName, and Price. Write R commands to: a) display the first five rows, b) display the last five rows.**
    [U] / [CO2]
    (2 marks for creating a data frame, 1.5 marks for display first five rows and 1.5 marks for display last five rows)

    Create data frame:

    ```
    product <- data.frame(ProductID, ProductName, Price)
    ```

    Display first five rows:

    ```
    head(product, 5)
    ```

    Display last five rows:

    ```
    tail(product, 5)
    ```

11. **Compare median and mode in real-life decision making with suitable examples.** **[An] / [CO3]**

    (1.5 Marks for the definition of Median and 1 mark for example, 1.5 Marks for the definition of Mode and 1 mark for example)

    Median – Middle value, Used for income data

    Mode – Most frequent value, Used for popular product size

Both support decisions

12. **What is the use of the str() function? Explain with an example. [U] / [CO2]**

(3 Marks – Definition, 2 Marks – Example)

Shows structure of object, displays data type, shows number of variables

Example: str(product)

Used to understand data

Part C

Essay Questions Answer any **1** questions

Each question carries **15** marks

13. **Explain the concept of row binding in R using rbind().Also Create two data frames with identical column names representing sales data of two different months. Combine them using rbind() [U] / [CO2]**

(Definition – 3 Marks; Key Points / Concept – 4 Marks; Syntax – 2 Marks; Example Creation – 3 Marks; Combining using rbind() – 3 Marks)

Row binding is the process of **combining data frames by adding rows one below another using rbind()**.

It Combines rows, requires same column names and number of columns, and returns one combined data frame.

Syntax: rbind(dataframe1, dataframe2)

Example Data Frames Creation:

jan <- data.frame(ProductID=c(1,2), Sales=c(5000,6000)); feb <- data.frame(ProductID=c(3,4), Sales=c(5500,6500))

Combining Data Frames Using rbind():

total <- rbind(jan, feb)

14. **With suitable R code, explain how to create and customize a pie chart using base R. [A] / [CO3]**

(Definition of Pie Chart – 3 Marks; Syntax of pie() – 2 Marks; Creating Data – 3 Marks; Customizing (labels, colors, title) – 4 Marks; R Code / Output – 3 Marks)

A pie chart is a graphical representation that shows data as proportions of a whole.

Syntax : pie(x, labels, main, col)

Creating Data :

sales <- c(40, 30, 20, 10)

labels <- c("Product A", "Product B", "Product C", "Product D")

Customizing (labels, colors, title)

Labels name categories, colors improve appearance, and title describes the chart.

R Code / Output

```
pie(sales,
    labels = labels,
    main = "Sales Distribution",
    col = c("red", "blue", "green", "yellow"))
```